In the Name of God, the Creator of All Crafts

Management Information Systems (MIS) Project

Phase 3: Executive Dashboard Design & Business Intelligence in Power BI

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1 Phase 3 Objective: Executive Visual Analytics

The culminating objective of Phase 3 is to master the design, engineering, and interpretation of enterprise management dashboards utilizing **Microsoft Power BI Desktop**. Transforming raw and structured database tables into real-time decision-support assets, students will model business analytics, extract critical operational KPIs, and evaluate manufacturing performance within the copper production facility.

2 Data Pipeline Refinement & Semantic Data Modeling

1. **Data Population & Logical Schema Integrity:**
   - Prior to building analytical visuals, populate database tables with sufficient operational volume to ensure resulting dashboards convey statistically meaningful and realistic patterns.
   - Students should refine and extend their Phase 2 relational schema as necessary, incorporating the provided course auxiliary datasets: the `Phase 2 Auxiliary Dataset` and `Mis_DataSet_Phase3`.
2. **Data Model Relationships (Star/Snowflake Schema Validation):**
   - Rigorously inspect relationship cardinalities ($1:1$, $1:N$, $M:N$) and cross-filter directions inside the Power BI Relationship Model view.
   - Rectify orphan tables, ambiguous pathways, and inactive relationships. Correct structural relationships are a strict prerequisite for precise DAX context transition and error-free visual filtering.
3. **Software Environment Compatibility:**
   - Instructional video modules for this phase were developed using **Power BI Desktop (May 2023 release)**.
   - To avoid UI discrepancies, ribbon differences, or feature incompatibilities, utilizing this specific version or a proximate release is strongly advised.

3 Executive Dashboard Architecture & Visualization Standards

Students are required to architect an intuitive, professional, and insight-driven business intelligence suite meeting the following minimum criteria:

Core Visualization Deliverables

- **Dashboard Sheet Count:** Design and deploy **at least three (3) distinct dashboard sheets (pages)**, each dedicated to a coherent operational domain (e.g., Comminution & Energy, Inventory Management, Quality Assurance).
- **Visual Density & Diversity:** Every dashboard page must contain **at least three (3) diverse visual elements**, including column/bar charts, donut/pie charts, tabular grids, matrix tables, numeric KPI cards, and trend line charts.
- **Analytical Commentary:** Every chart, trend visual, and performance metric must be accompanied by concise analytical notes directly within the report, explaining the operational insight conveyed by the data.

4 Advanced Analytics via Power Query (M) and DAX

Leverage **Power Query** for extract-transform-load (ETL) routines and **Data Analysis Expressions (DAX)** to author **at least three (3) advanced analytical measures, calculated columns, or data structures**:

- **Complex Calculated Columns:** Conditional scoring, multi-column logic, or nested classification flags.
- **Auxiliary & Dimension Tables:** Creating supporting tables such as automated Date/Calendar tables (`CALENDARAUTO()`) or custom category grouping tables.
- **Specialized DAX Business Measures:** Implementing time-intelligence metrics, moving averages, target fulfillment percentages (% Target Realized), Month-over-Month (MoM) growth variance, or dynamic ranking.

Requirement: All authored DAX measures and custom calculated columns must be actively integrated into dashboard visuals, serving an explicit analytical purpose.

5 Key Performance Indicators (KPIs) & Enterprise Analysis

Strategic Plant KPI Modeling

Based on the copper manufacturing scenario and operational telemetry, formulate and visualize **at least two (2) mission-critical Key Performance Indicators (KPIs)**.

- **Advanced Visualization:** Deploy sophisticated visuals such as Radial Gauge visualizers, KPI cards with status thresholds, target-vs-actual bullet/combo charts, or cumulative recovery trend curves.
- **Executive Narrative Assessment:** Formulate a multi-sentence diagnostic critique for each KPI, explicitly answering:
  - Does current performance reflect positive operational health or warrant strategic intervention?
  - What underlying operational variables (e.g., machine downtime, ore hardness, reagent consumption) drive the observed outcome?

6 Final Submission Package Format

Deliverables must be submitted via the course portal as a single compressed **ZIP** archive structured as follows:

1. **Native Power BI Project File:** Complete, uncorrupted project file with extension `.pbix`.
2. **High-Resolution Visual Captures:** Clear, full-screen captures of all three designed dashboard sheets.
3. **Formula & Code Repository:** Text file or annotated code captures documenting all custom DAX measures and Power Query M-code transformations.
4. **Analytical PDF Report:** Complete technical and diagnostic commentary explaining all visual components, trends, and modeled KPIs.
5. **Consolidated PDF Document:** A single master PDF document compiling all items above as the formal final executive report.